

BU5300 – IT Project Management

Lecture 3: Development Lifecycles

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Today's Agenda

- Quick review of Lecture 2
- Chapter 6: Development Lifecycles and Approaches
 - Traditional approaches
 - Agile approaches

Takeaways from Lecture 2

- Building a business case to obtain funding
- The innovation cycle
- Start-up and Grants for Student Entrepreneurship
- Project costs (one-time vs recurring)
- Benefit analysis
- The organizational framework, key roles and responsibilities
- The project structure
- Sponsor and project manager

Last week's assignment

Build a **Business Case document (or presentation)** for your IS project idea

- Introduction and background
- Management summary
 - Description of problem or opportunity
 - Options available and considered
- Cost/benefit analysis
- Impacts and risks
- Conclusions and recommendations

Tip! You can use the PRINCE's template available at:

<https://kristiania.instructure.com/files/1054531>

Options Considered

VisitOslo's STN example

- Hire more [developers, designers, etc]
 - Faster development cycle
 - Faster market entry
 - Higher initial costs
 - Larger management overhead
- Paying fee for downloading app
 - Immediate revenue
 - Could deter potential users
 - Require more aggressive marketing
- Continue business as usual
 - No immediate financial investment
 - Loss of competitive edge
 - Risk of stagnation
- Expanding on current platforms (ex. VisitOslo website)
 - Builds on established user base
 - Utilizes existing infrastructure
 - Miss new customer segments
 - Need to update project strategy

Cost Analysis

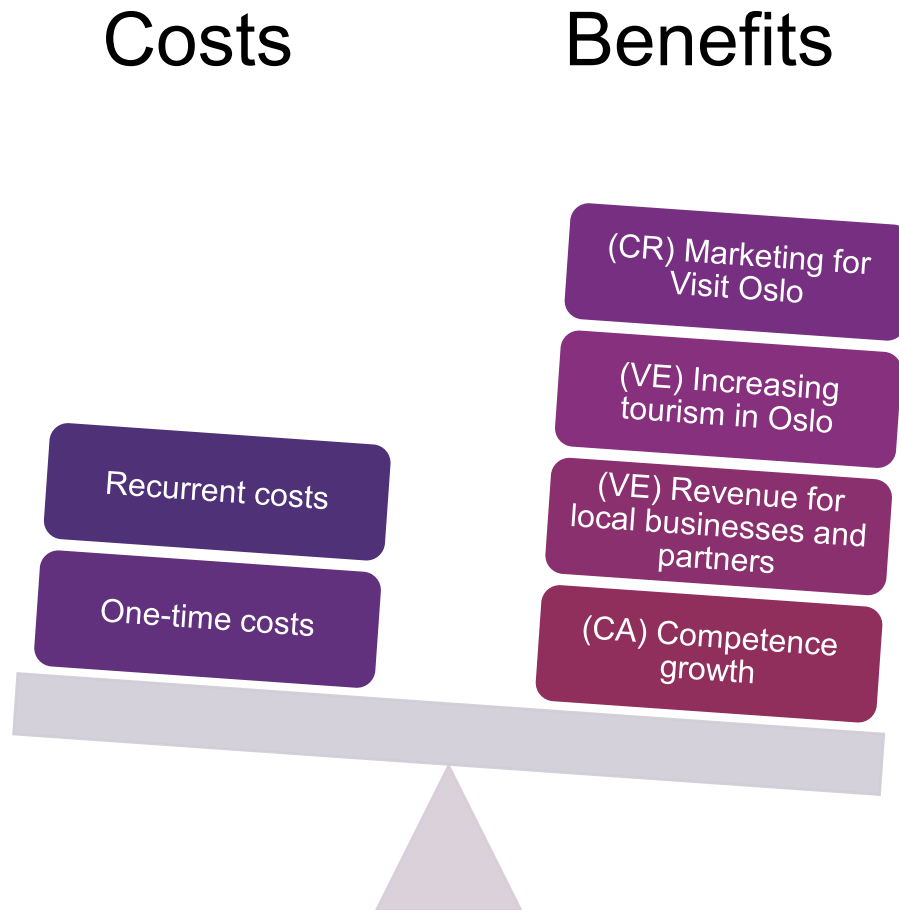
One-time (first-year development)

- Purchase of hardware and software
 - One-time licenses
- Salaries of IT, employees, consultants involved in the project development (before release)
- Office set-up (furniture, telephones and other equipment, etc)

Recurring Costs (operations)

- IT infrastructure costs
 - Yearly licenses (for maintenance)
- Salaries of IT, employees, consultants involved in the project maintenance (after release)
- Office maintenance (electricity, internet provider, insurance, etc)
- Technical support

Benefit Analysis



How to describe benefits:

- Payback calculation (or discount cash flow)
- Return of Investment
- Cause and effect

Legend:

(CR) Cost reduction

(CA) Cost avoidance

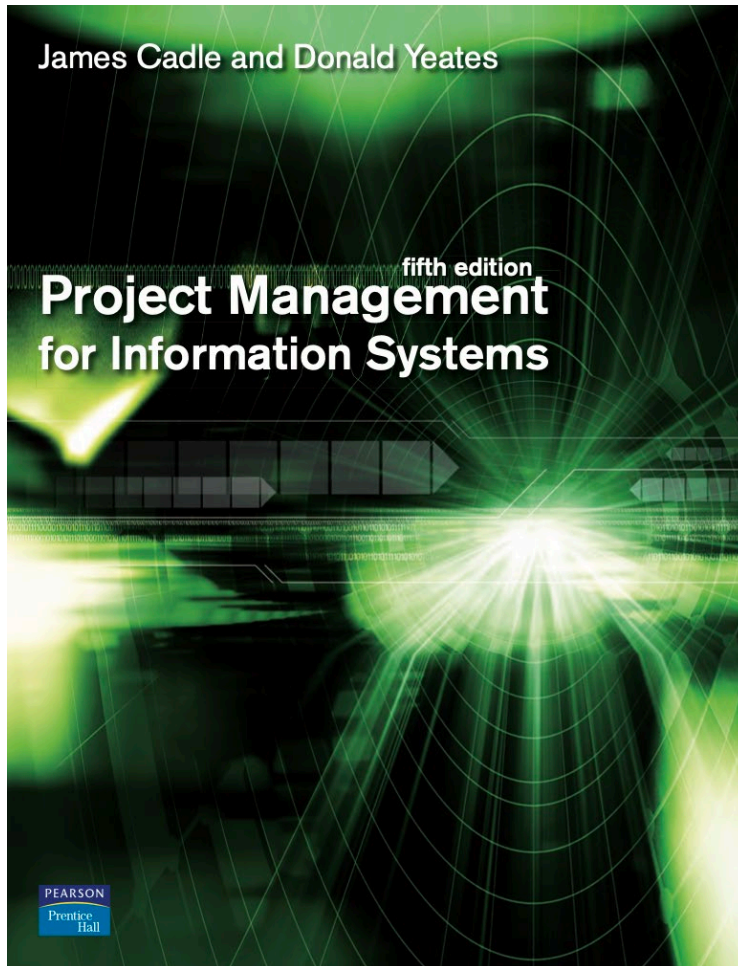
(VE) Value enhancement

Risk Analysis

Risks	Probability	Impact
Technology change, e.g., Android and iOS support	?	?
Business change (internal or external)	?	?
Unexpected or unbudgeted expenses	?	?
Similar app released by a competitor	?	?
Development uncertainties leading to delays	?	?
Cyber attacks to our database	?	?

Overview of the Course

Lectures and Book Chapters



Lecture 1 - Chapters 1 and 2

Lecture 2 - Chapters 3 and 4

Lecture 3 - Chapter 6

Lecture 4 - *Agile Development*

Lecture 5 - Chapters 7 and 8

Lecture 6 - Chapter 9 and 10

Lecture 7 - Chapters 11, 12 and 13

Lecture 8 - Chapters 14 and 15

Lecture 9 - Chapters 18 and 21

Lecture 10 - Chapters 20, 22 and 23

IMPORTANT! Changes may occur. Always check on Canvas and TimeEdit

Chapter 6: Development Lifecycles and Approaches

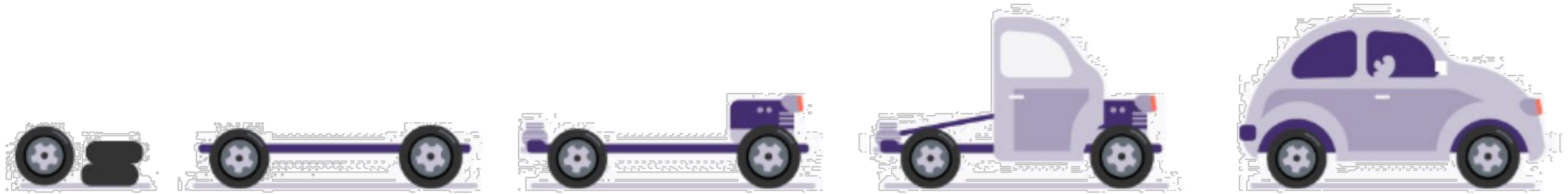
Learning Outcomes

- Identify different development lifecycle models
- List business and technical reasons for choosing a development lifecycle
- Contrast ‘traditional’ and agile systems development
- Describe principles of the object-oriented and component-based approaches
- Suggest an approach that addresses the uncertainty of real-world situations

- The software development lifecycle (SDLC) is a streamlined process / structured framework for creating software
 - **Time**-efficient and **cost**-effective
 - Aligned with specific goals, meeting business case and customer expectations (**scope**)
 - Minimize risks
- Why?
 - A project is defined as something that has an end
 - The objective of the project may cover only the delivery of a specified product
 - The project lifecycle covers the delivery of the end-product of the project
 - The system development lifecycle often covers only the technical deliverables

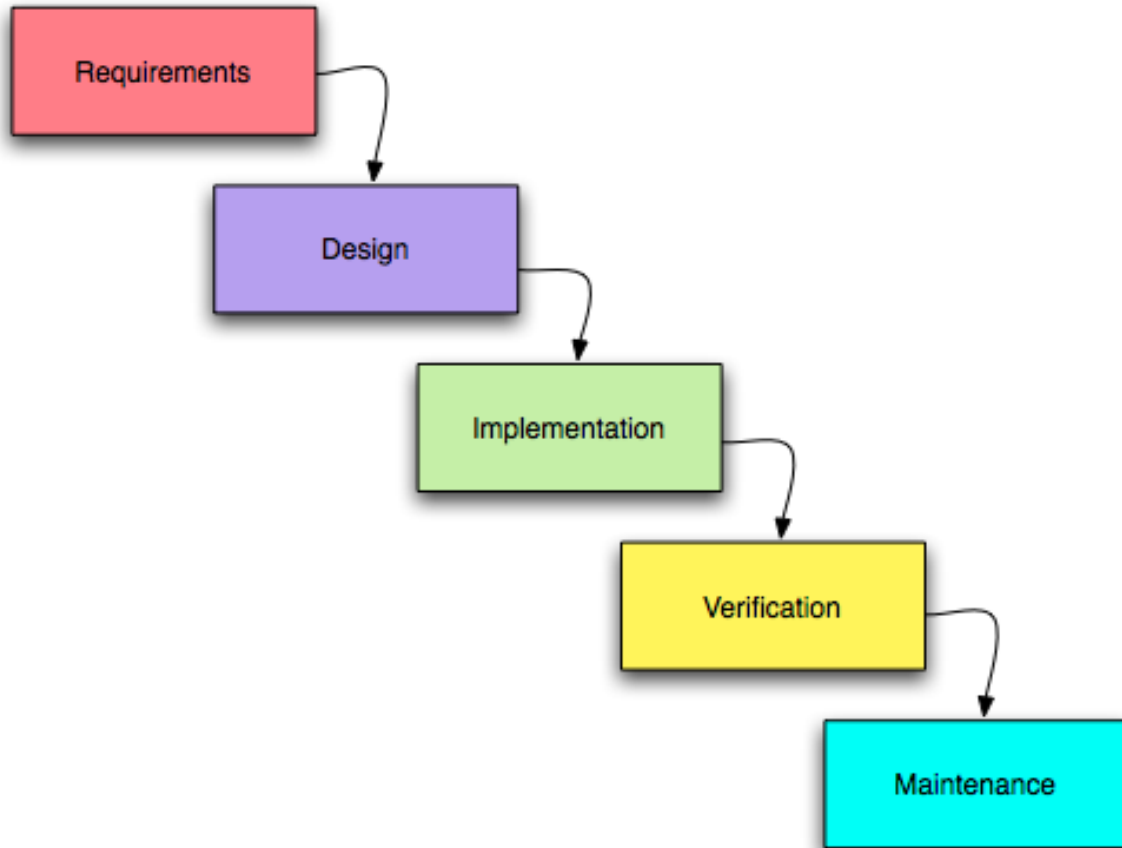
Example of Project SDLC

Business Case



Waterfall model

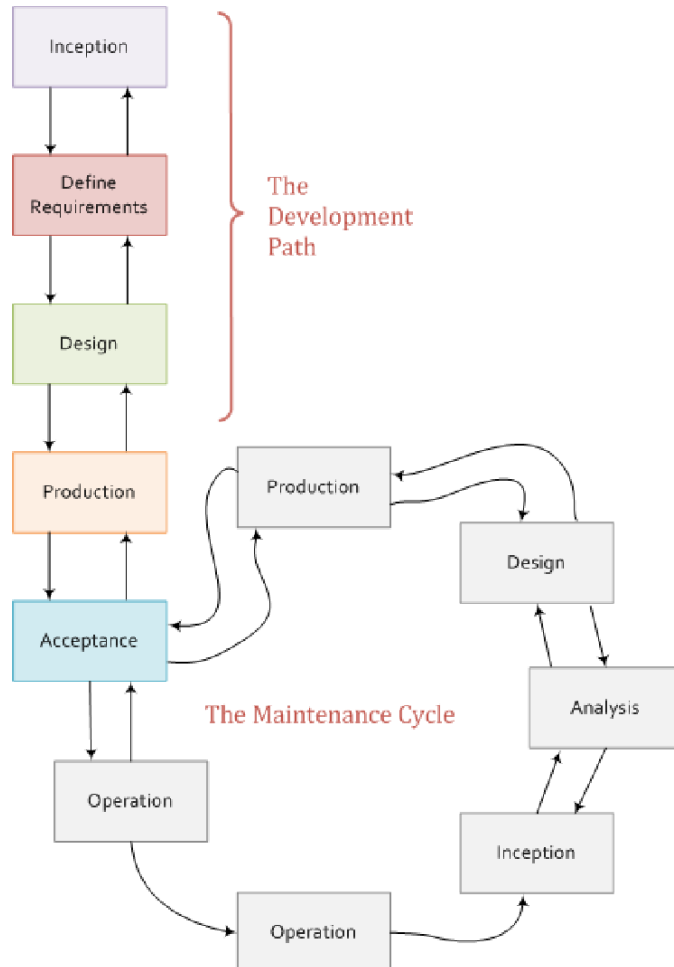
System development lifecycle



- Breakdown of project activities into linear sequential phases
- Progress flows in one direction, so it lacks iteration and flexibility
- There is some degree of iteration of work and products within a stage but very little between stages

The 'b' model

System development lifecycle

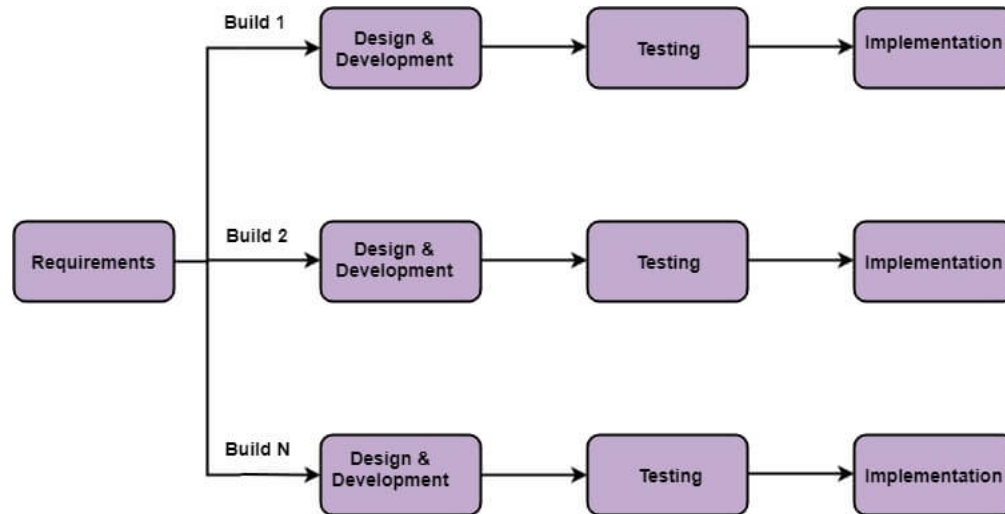


- Extends the waterfall model
- Addresses weaknesses in the maintenance phase
- Create an evolutionary enhancement process (see also spiral model)

N. D. Birrell and M A Ould, *A Practical Handbook for Software Development*, Cambridge University Press, 1985.

The incremental approach

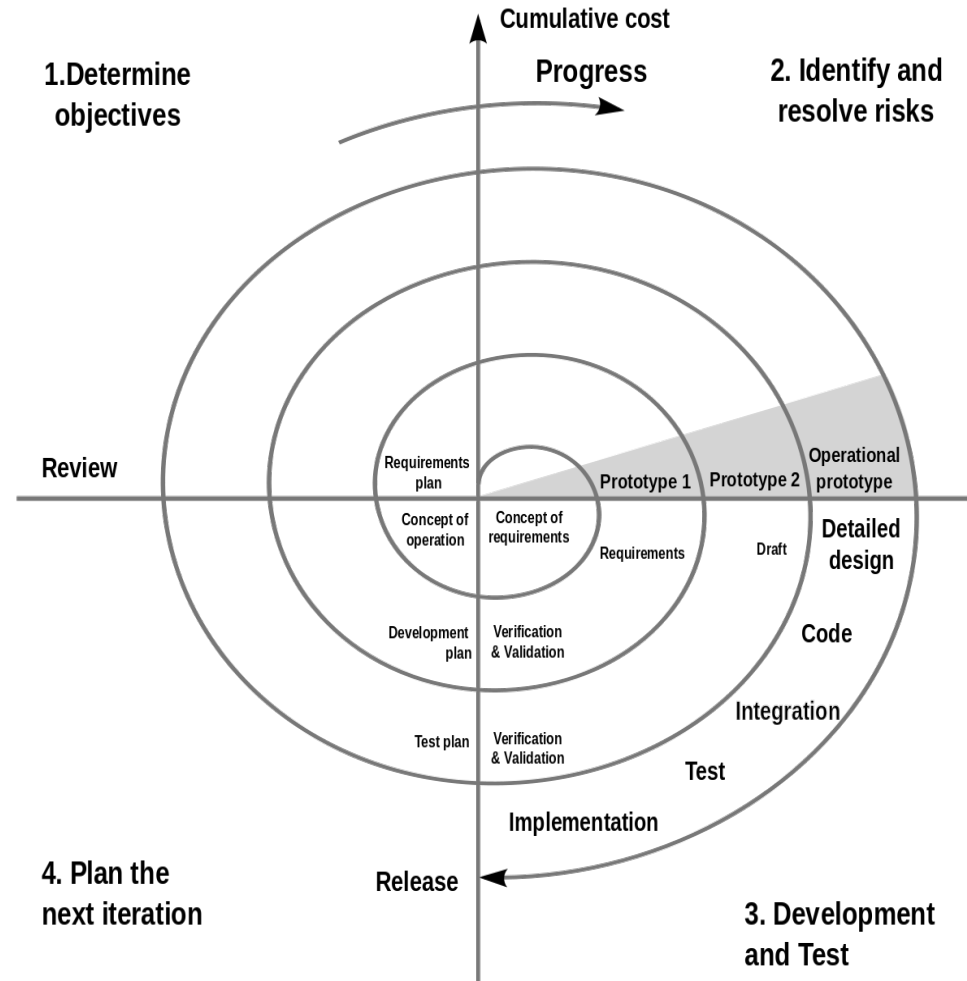
System development lifecycle



- Another variant of the waterfall model
- Used where the total system is to be delivered in phases
- Makes the delivery and testing more manageable
- Introduces the new system over time, allowing familiarization with the changes

The spiral model – evolutionary development

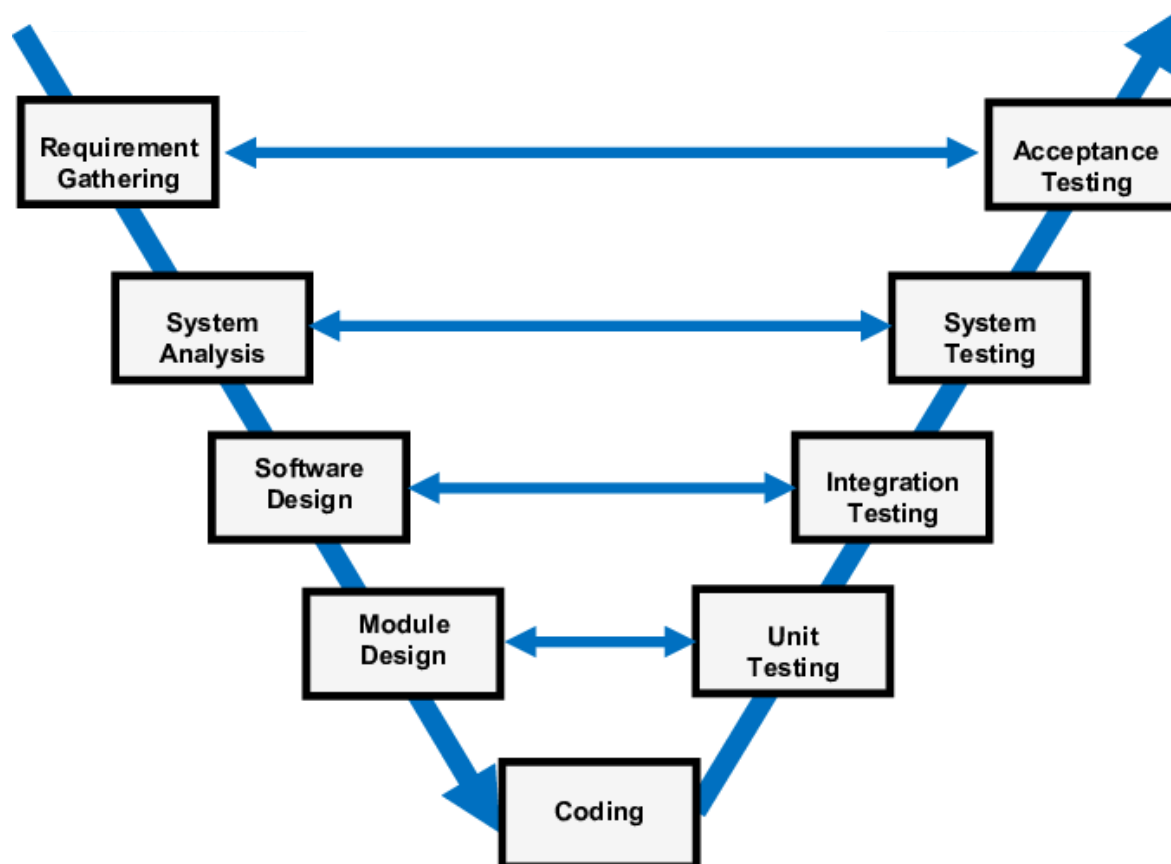
System development lifecycle



Boehm, Barry W. **A spiral model of software development and enhancement.** *Computer* 21.5 (1988): 61-72.

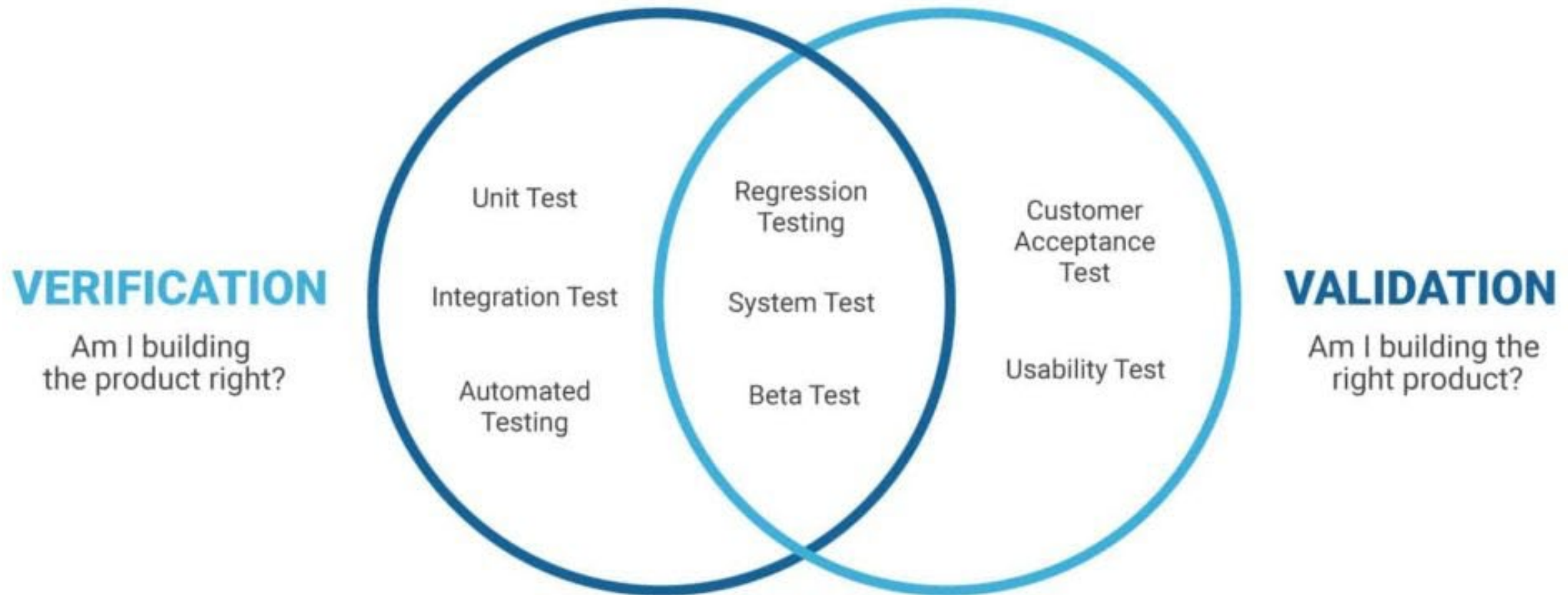
The 'V' model

System development lifecycle



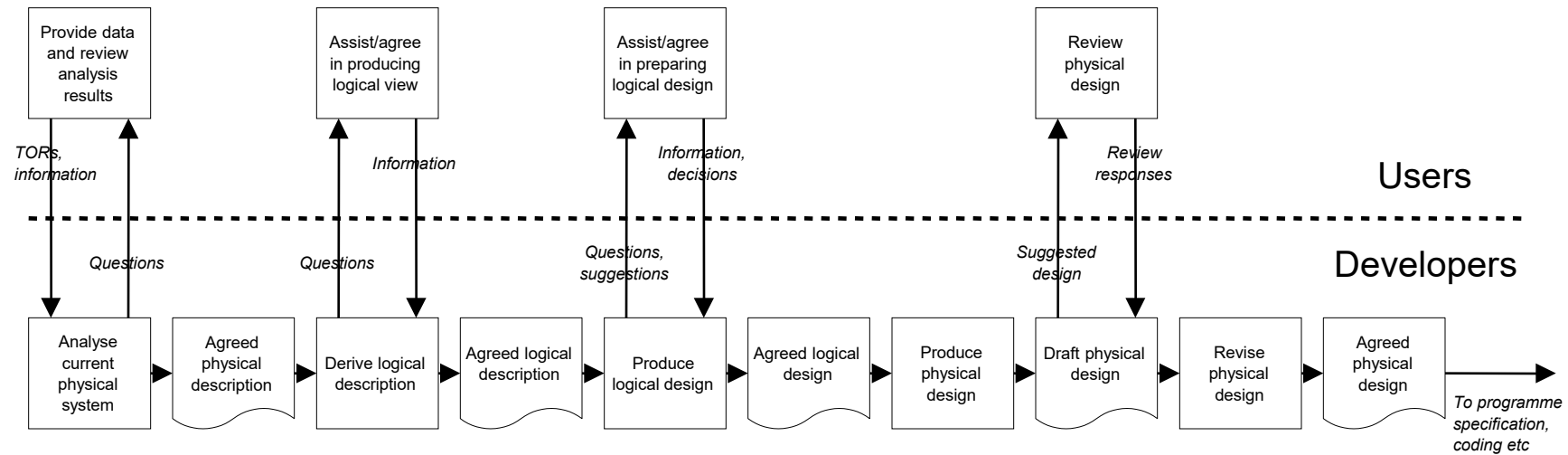
- Another variation of the waterfall model
- The left, downward leg shows the project definition, increasing breakdown of the system
- The right, upward leg shows the integration and testing, culminating in the delivered product
- Focus product quality by verification and validation
- Correspondence between the different stages in the project

Verification and Validation (V&V)



Structured Systems Analysis and Design Method

System development lifecycle



Greater user involvement at all stages; driven by users

Agile Methodology

The Agile methodology is a project management approach that involves breaking the project into phases and emphasizes continuous collaboration and improvement.

Teams follow a cycle of planning, executing, and evaluating.

Agile approaches – features

- Success dependent on empowerment of users and developers
- Deliverables reviewed for business fitness
- Testing integral to iterative lifecycle
- All changes reversible
- Incremental/partial delivery acceptable
- ‘Timeboxing’ used to control timescale (and budget)
- Workshops widely used

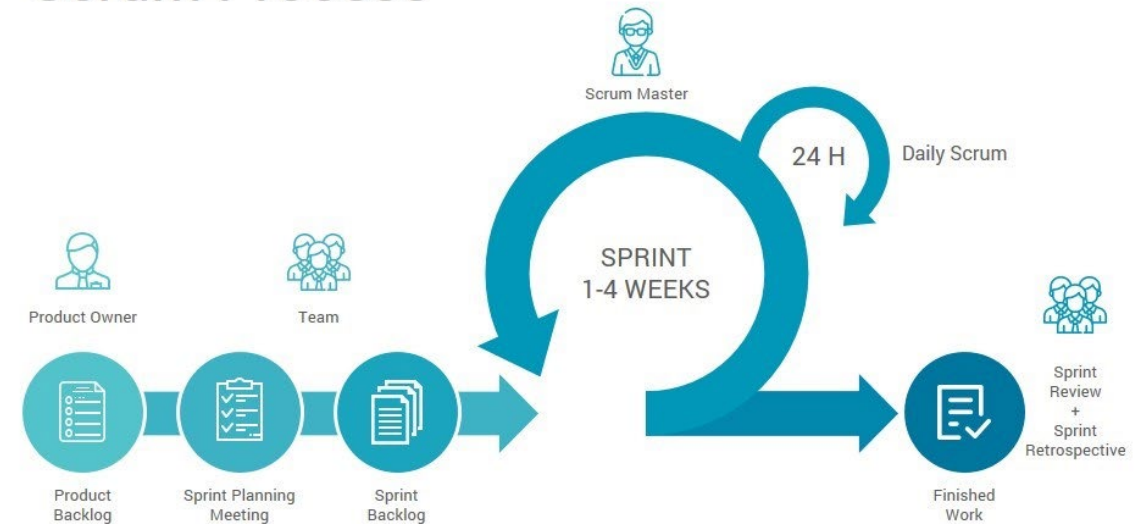


Scrum

Agile methods

- Scrum is daily meeting to review progress and reset priorities
- Development done in 'Sprints'
- Every aspect of the project time-boxed
- Project manager is 'ScrumMaster' – different from the usual PM role
- Teams self-governing

Scrum Process

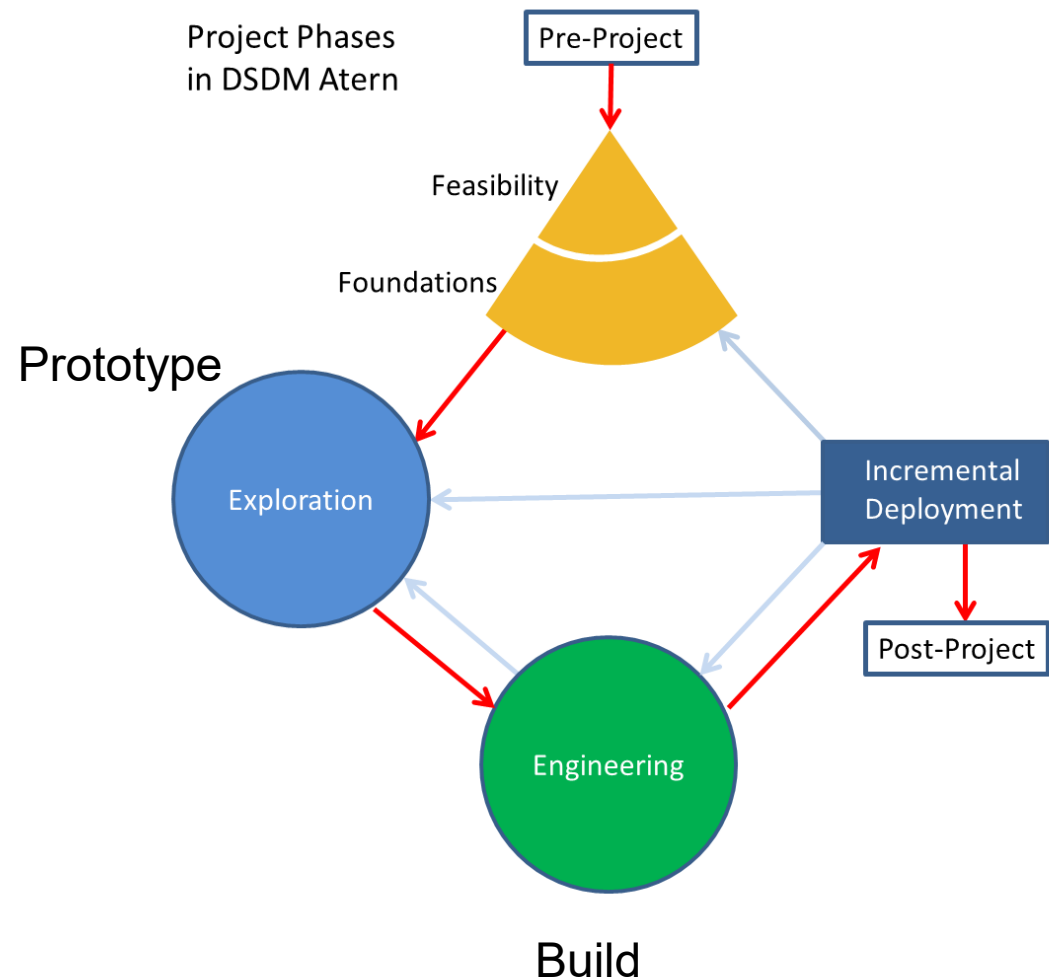


DSDM (Dynamic System Development Method)

Agile methods

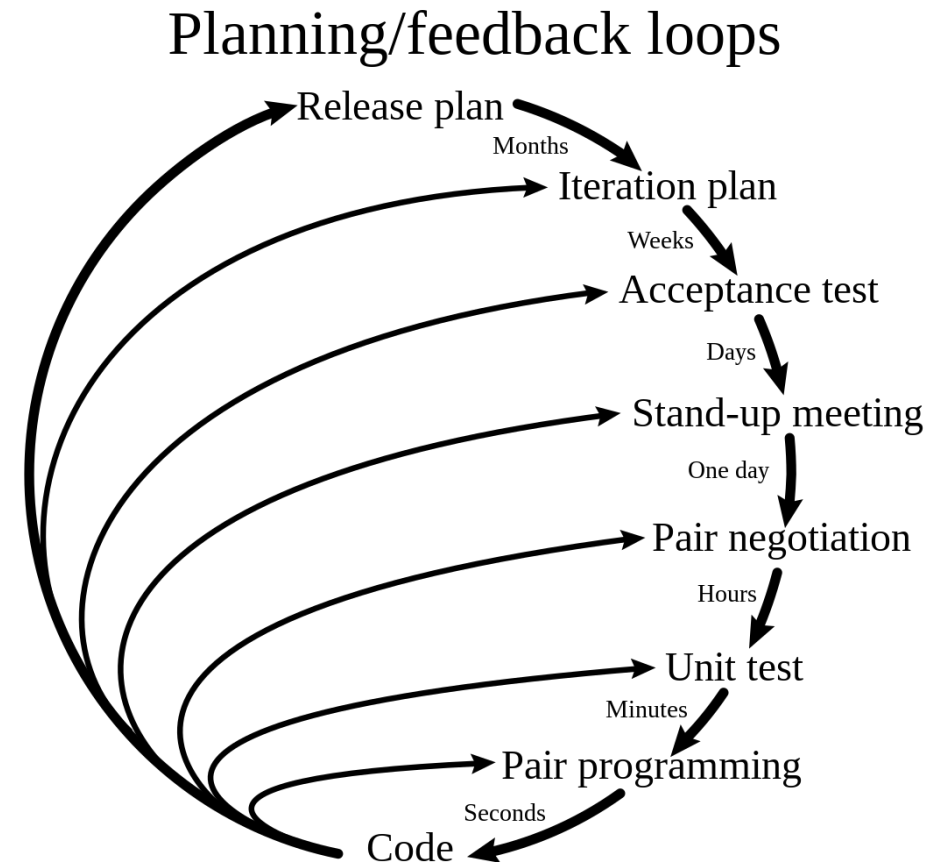
Eight principles:

- 1 Focus on business need
- 2 Deliver on time (Tight Deadline)
- 3 Collaborate
- 4 Never compromise quality
- 5 Develop solution iteratively
- 6 Build incrementally from firm foundations
- 7 Communicate continuously and clearly
- 8 Demonstrate control



Extreme programming

- Deal with rapidly changing requirements
- Works best on relatively small projects
 - Or enhancements to existing systems
- Developers work in pairs
- Testing – integral part of process
- Emphasis on frequent releases of small-scale packages of software

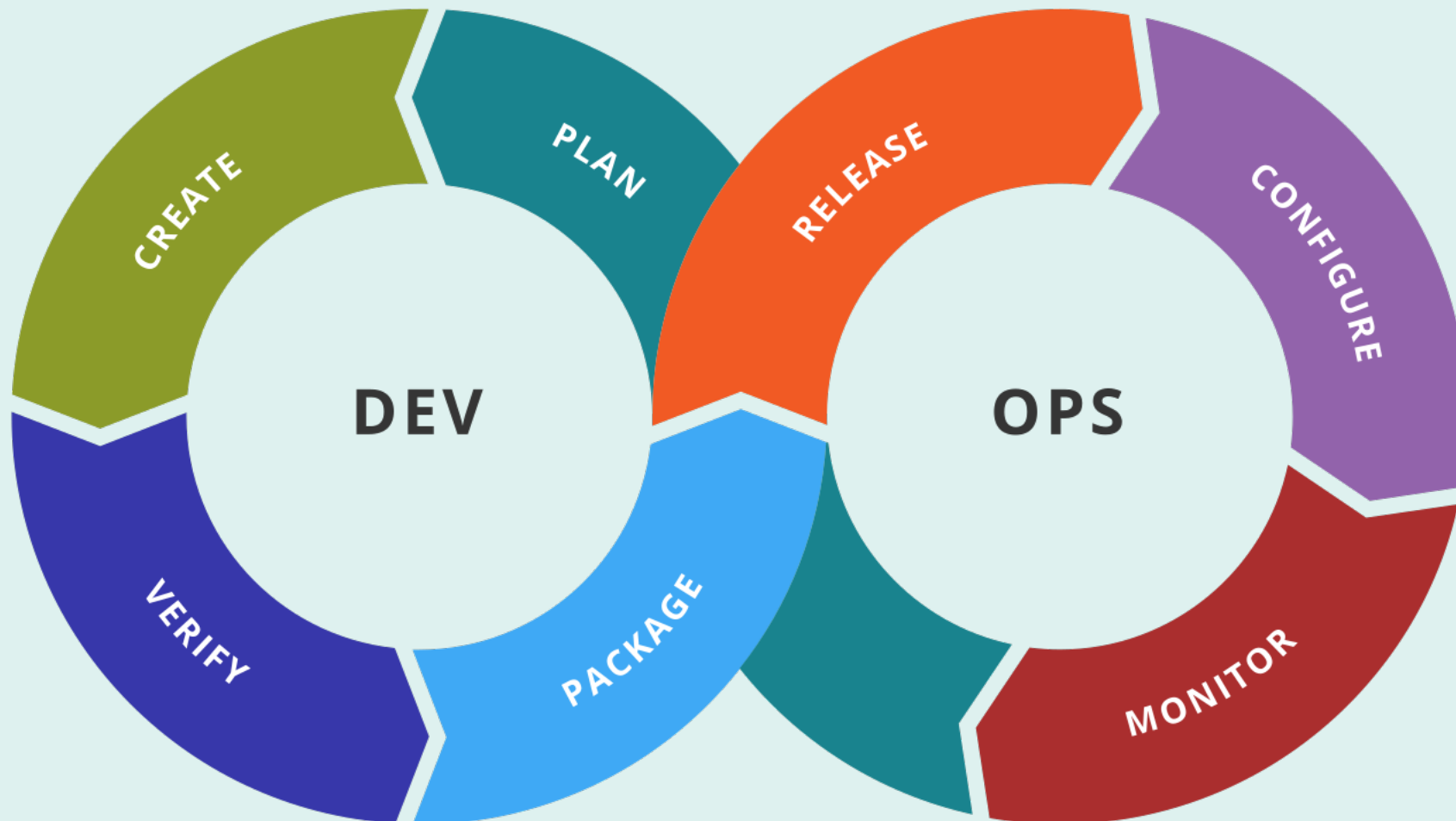


Sidenote:

Complementary paradigms / techniques

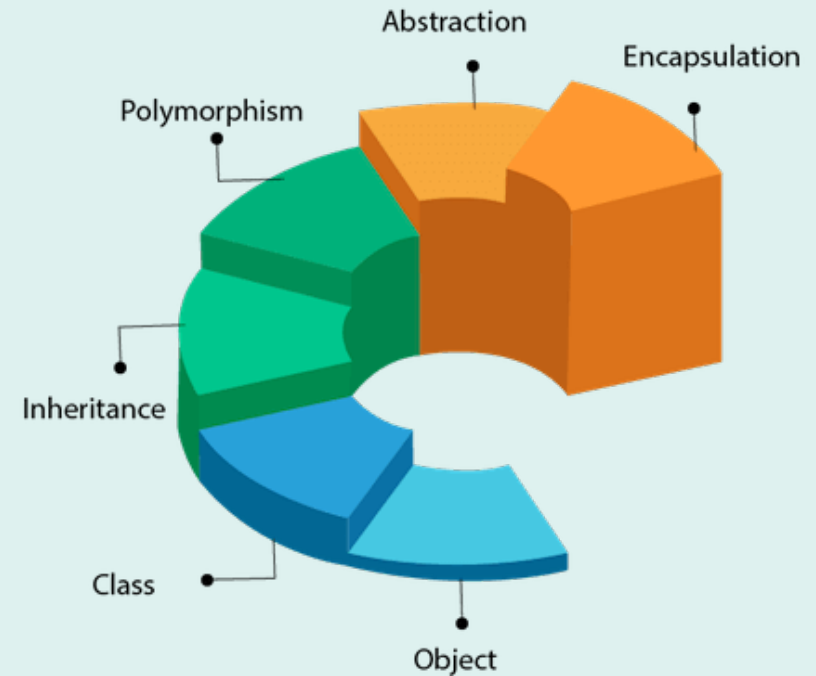
not lifecycles, more like an extension

DevOps



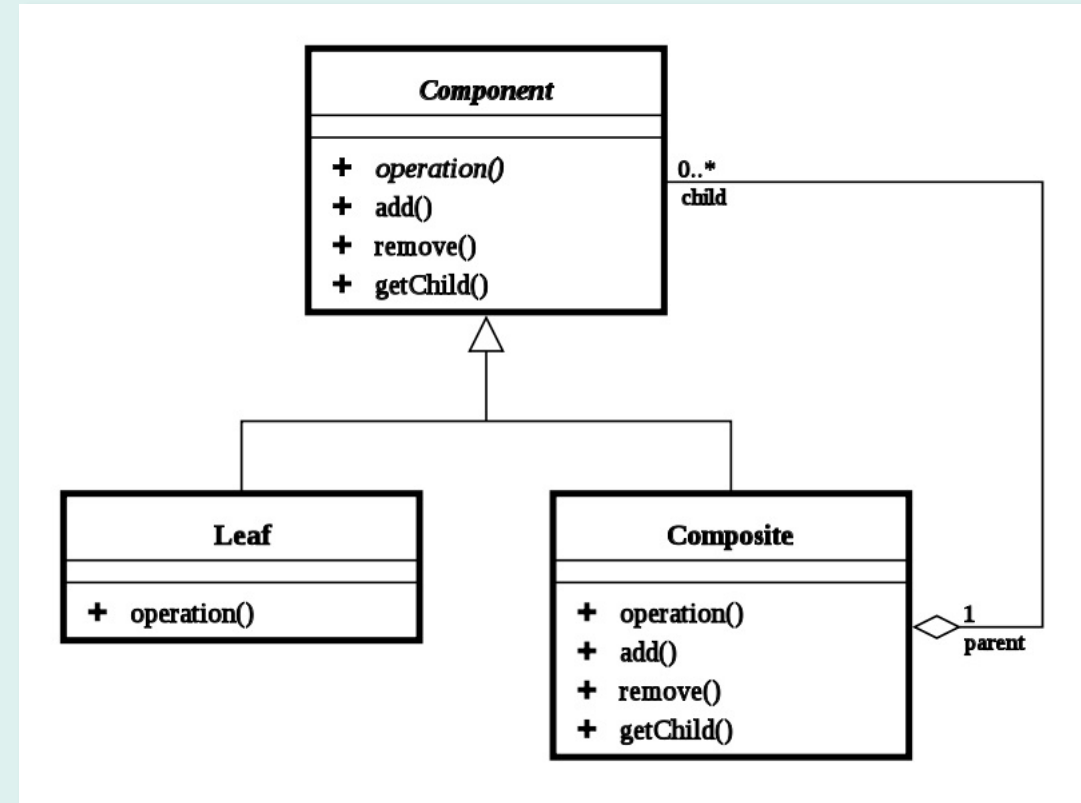
Object-oriented development

- Object is a package of software containing:
 - 'variables' (data)
 - 'methods' (processes)
- Objects communicate via messages
- System is built up from intercommunicating objects
- Deals with problem of integration of large systems



UML and Unified Process

- Unified Modelling Language provides visual language for OO projects
- Approach 'open' (non-proprietary) but Rational Corporation offers Rational Unified Process
- Four phases to process:
 - Inception
 - Elaboration
 - Construction
 - Transition



Component-based development

- Development of reusable components
- Aim – to create libraries of components that can be combined to build new systems
- Long-term benefits – reduce development costs and produce more reliable systems
- Short-term costs often higher – because of need to consider wider usage of components

Package-based IS projects

- Buying commercial off-the-shelf (COTS) is quicker than build from scratch
- Two main types of project:
 - Package-constrained – users adapt to what package can do
 - Package-based – package tailored to users' exact needs
- Extensive tailoring probably more expensive than development
- The more the change in the business processes the more the change required on the people side
- See chp.20 of the textbook - Organizational Change Management

Questions?

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Class Activity

- For the IS project discussed in the last lecture, select a development lifecycle model that best suits its characteristics.
- Justify your choice using key points from Chapter 6.
- Compare your chosen lifecycle model with other relevant models (a few selected ones) to explain why it is the most appropriate for this project.
- Write your response as a short memorandum to the project sponsor, justifying your choice and reasoning.

Class Activity:

Development Lifecycles and Approaches

Go to

www.menti.com

Enter the code

5724 7335



Or use QR code

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- Ruhe, Günther, and Claes Wohlin, eds. **Software project management in a changing world**. Vol. 21. Berlin/Heidelberg, Germany: Springer, 2014.
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